Perpendicular offset from the mirror line is bimodal for height, unimodal for BMI

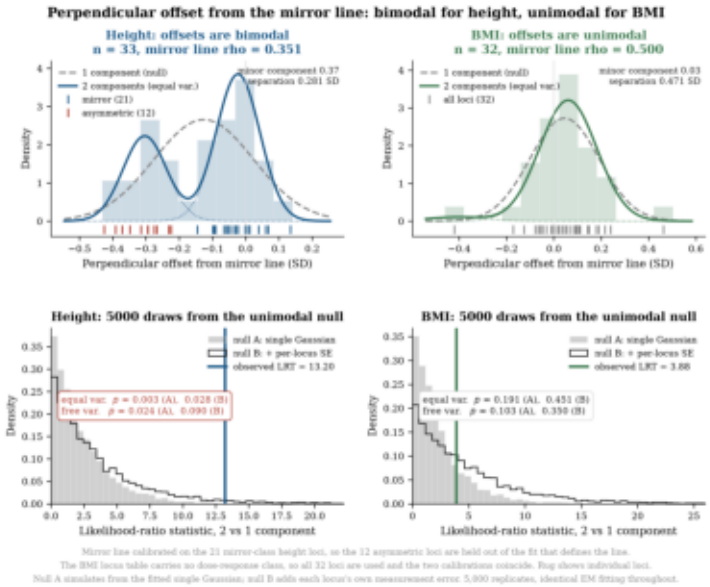
Exact Sacks reciprocal-CNV effects | parametric bootstrap, 5,000 replicates | 2026-10-10

Methods

The mirror model places every locus on the line DUP = -rho*DEL, with position along the line set by gene count. Each locus's signed perpendicular distance from that line, offset = (rho*DEL + DUP)/sqrt(1+rho^2), is what the mirror model does not explain: a discrete asymmetric class should make it a two-component mixture, a continuous process a single Gaussian.

One- and two-component Gaussians were compared by likelihood ratio. Mixture likelihood ratios do not follow chi-square, so the null came from a parametric bootstrap: 5,000 datasets of the same size simulated from the fitted single Gaussian, both models refitted to each. Observed and simulated data were fitted with the identical EM procedure and the same 25 random starts, since fitting the observed data harder than the nulls biases p downward. Null A is the plain fitted Gaussian; null B adds each locus's own measurement error. The equal-variance alternative (2 extra parameters) is primary, being what a sparse hit implies: it displaces a locus without changing its dispersion. Free variance (3 parameters) is reported alongside.

The mirror line was calibrated two ways. Primary: on the 21 mirror-class height loci only, holding the 12 asymmetric loci out of the fit that defines the line (rho = 0.351). Comparison: on all 33 loci, where the asymmetric loci tilt the line their own displacement is measured against (rho flattens to 0.218). The BMI table carries no dose-response class, so all 32 loci are used and the calibrations coincide.



Offsets with fitted one- and two-component models (top); observed statistic against the bootstrap null (bottom). Rug = individual loci.

Results

Trait	Mirror fit	Alternative	LRT	Minor wt.	Separation (SD)	p (null A)	p (null B)
Height	mirror-class	equal var	13.20	0.37	0.281	0.003	0.028
Height	mirror-class	free var	13.20	0.37	0.280	0.024	0.090
Height	full data	equal var	5.22	0.38	0.291	0.103	0.158
Height	full data	free var	7.59	0.23	0.301	0.196	0.287
BMI	either	equal var	3.88	0.03	0.471	0.191	0.451
BMI	either	free var	9.40	0.03	0.482	0.103	0.350

Findings and caveats

- Height is bimodal: LRT 13.20, p = 0.003 (A) / 0.028 (B) equal-variance, 0.024 (A) free-variance. The minor component holds 37% of loci at 0.281 SD separation, matching the sparse single-hit displacement (~0.27-0.31 SD).
- BMI is unimodal throughout (p = 0.103-0.451). Its larger free-variance statistic (9.40) is an artefact of one locus pinned at the variance floor; equal variance removes it.
- The two height modes are exactly the published classes: asymmetric loci mean -0.306 SD, mirror loci -0.023 SD, all 12 asymmetric loci below -0.18 and none of the 21 mirror loci. Because the line is fitted on mirror-class loci and the labels themselves encode dose-response behaviour, this partly restates the labels rather than finding structure blind. The label-free version is the full-data calibration, which is not significant (p = 0.103-0.287); significance also weakens under null B. At 33 loci this is suggestive, not settled, and both numbers belong in a write-up.

Code, figure and tables

Analysis	`code/perp_offset_bimodality.py`	Figure	`figures/Figure_perp_offset_bimodality.{pdf,png}`
Figure script	`code/make_figure.py`	Results table	`outputs/bimodality_test.csv`
This report	`code/make_report.py`	Per-locus	`outputs/perp_offsets_per_locus.csv`
Full notes	`README.md`	Null draws	`outputs/null_LRT_draws.npz`

All paths relative to `...\Molly\ContinuousVsSingleHit\perp_offset_bimodality\` . Reproduce: `python code/perp_offset_bimodality.py --nboot 5000 --starts 25 && python code/make_figure.py && python code/make_report.py`